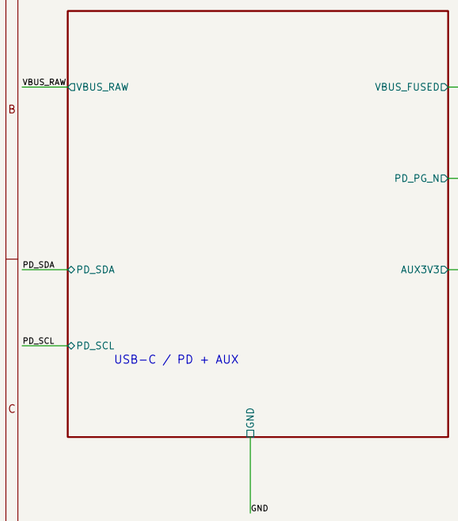
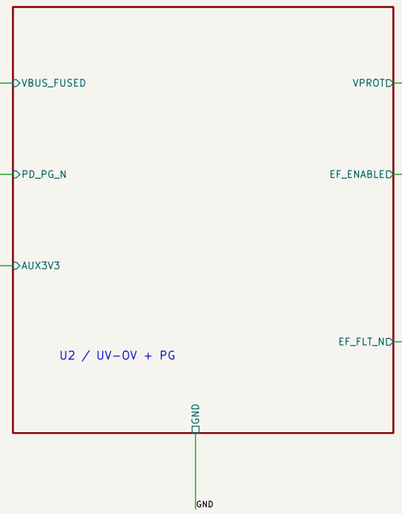
BIG MODULAR / WIRED POWER FLOW

REVIEW ONLY — NOT QUALIFIED FOR POWER-UP OR FABRICATION

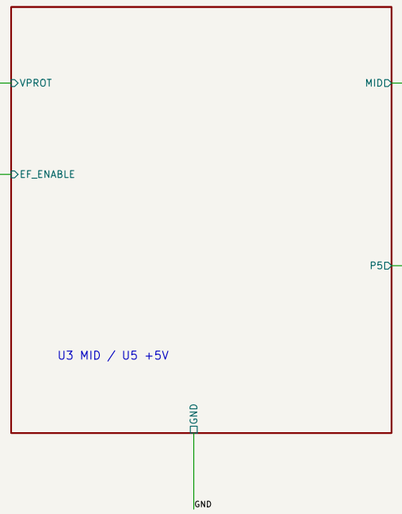
USB PD and always-on housekeeping



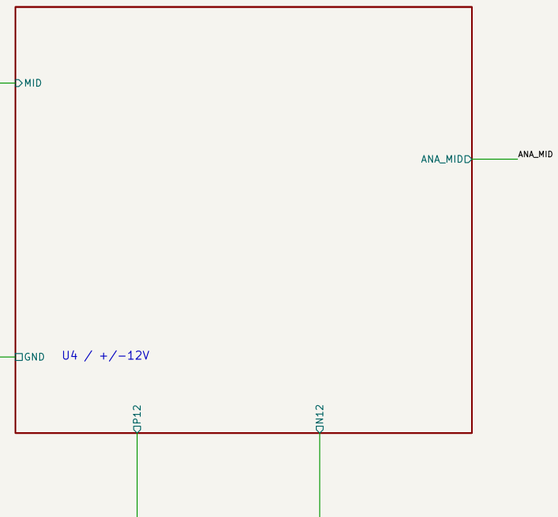
Independent hardware gate



Independent buck converters

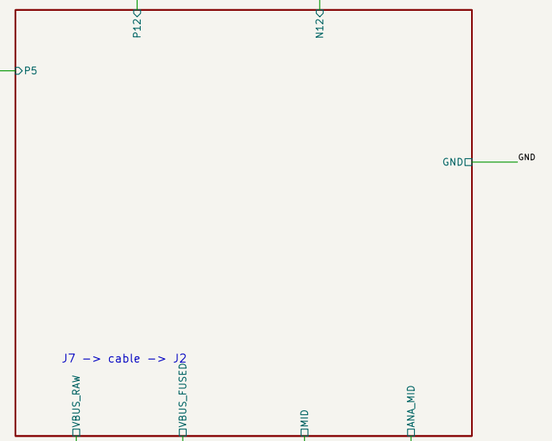


Direct bipolar rails

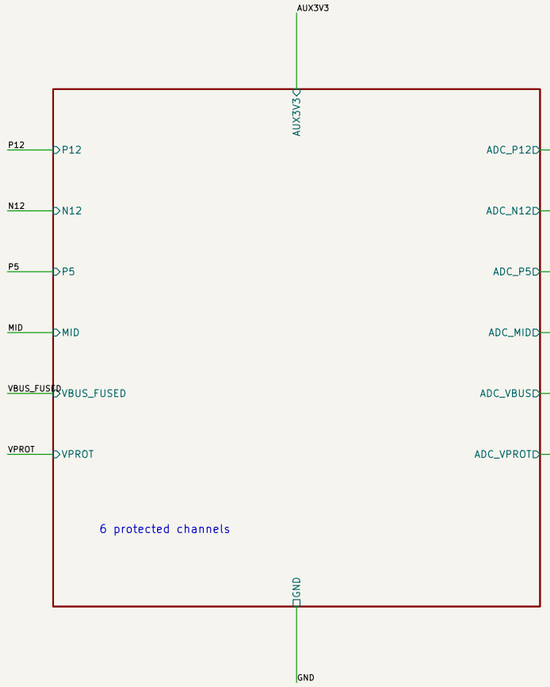


SENSE -> DEBUG: six protected analog outputs
J8 optional20-pin DEBUG ONLY; NOT POWER.
No MCU, blink or external enable dependency.

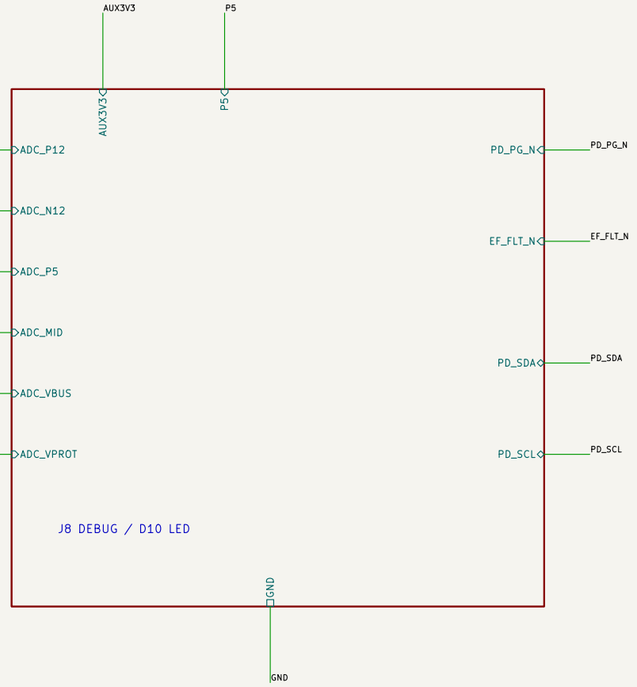
Manual output harness



Rail sensing and input protection



Rail debug and single power LED



J7 PCB landing -> one flying lead -> J2 keyed POWER OUT
Flag-only ports at bottom preserve independent ERC declarations.
RTN stays local to hardware gate; never substitute GND.

INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE

Big Modular

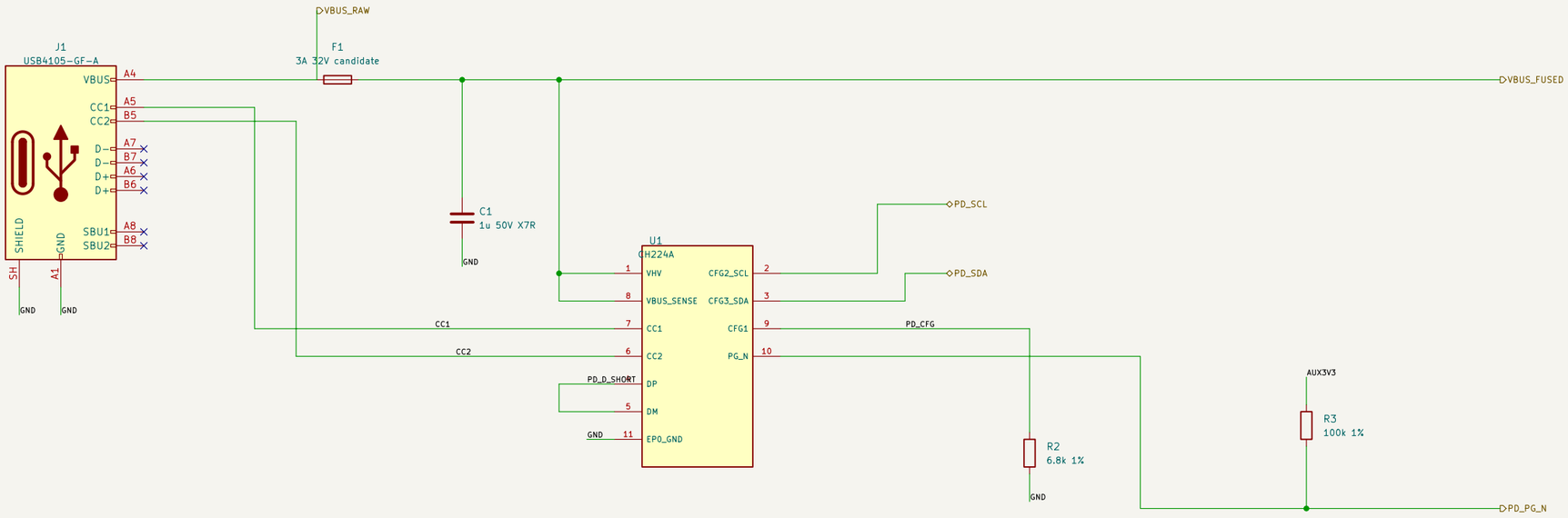
Sheet: /
File: big_modular.kicad_sch

Title: Big Modular — Electrical prototype / hierarchy

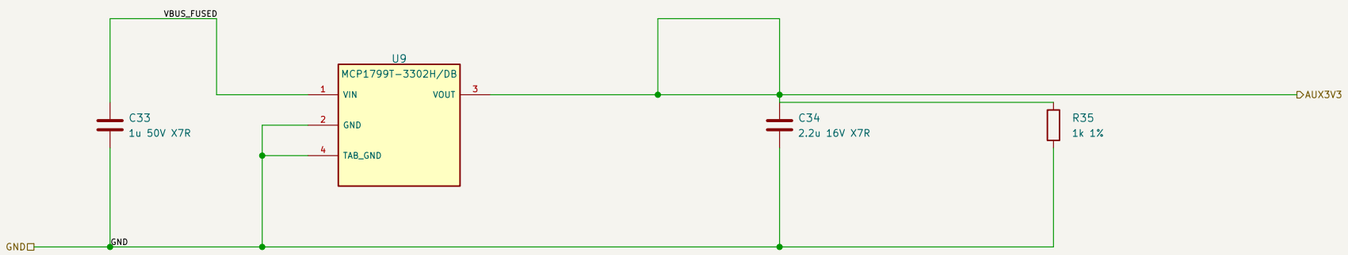
Size: User	Date:	**Rev: Wired review**
KiCad E.D.A. 10.0.6		Id: 1/8

01 / USB PD AND ALWAYS-ON HOUSEKEEPING

CH224A: VHV1 + VBUS8 directly fused VBUS: EP0 -> pad11.
WCH section 7.1 contradicts sections 4/6/8 on VBUS rating: vendor confirmation is a POWER-UP BLOCKER.
AUX3V3 operates at negotiation 5 V; NO output-rail operation from fallback. No mains/input = no LEDs.



Direct VBUS feed replaces the old CH224K 1k supply dropper.
I2C 0x22/0x23; status 0x09 bit3 = PD active; current 0x50 x50mA.
PG alone is not proof. Verify WCH silicon/document discrepancy before power.



INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE

Big Modular

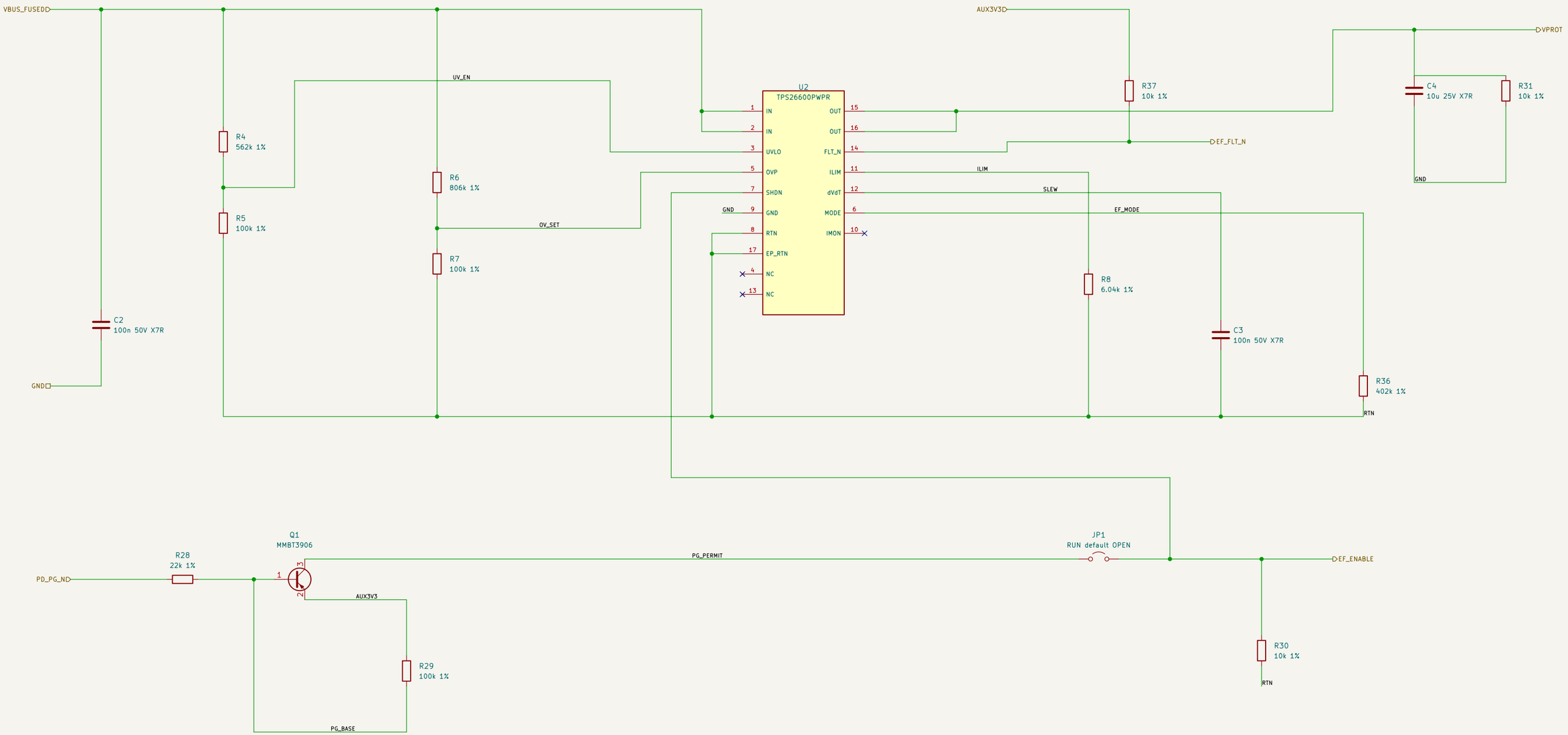
Sheet: /USB PD and always-on housekeeping/
File: 01_input_protection.kicad_sch

Title: Big Modular — USB PD and always-on housekeeping

Size: User	Date:	**Rev: Wired review**
KiCad E.D.A. 10.0.6		Id: 2/8

02 / INDEPENDENT HARDWARE GATE

7.878 / 10.781 V nominal hardware window. RTN is NOT GND.
R8 6.04k: 1.987 A nominal, not 3 A. MODE 402k: current limit with thermal latch-off.
JP1 RUN default OPEN: PG hardware permission; no debug controller required. FLT is aggregate, not per-rail overload proof.



Hardware PG -> Q1 -> JP1 -> SHDN, always subject to UV/OV window.
RUN must be deliberately closed only after power-up blockers are cleared.

INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE

Big Modular

Sheet: /Independent hardware gate/
File: 02_hardgate.kicad_sch

Title: Big Modular — Independent hardware gate

Size: User	Date:	**Rev: Wired review**
KiCad E.D.A. 10.0.6		Id: 3/8

03 / INDEPENDENT BUCK CONVERTERS

U3 MID 5.216 V: upper DC corner 5.357 V BEFORE transients. U4 ceiling 5.5 V.
U5 direct 5.011 V from VPROT, never from positive boost, 100n BOOT-to-SW each.
Effective ceramic C, magnetic Isat/DCR/thermal and loop/transient response remain qualification gates.

U3: MID / 5.216 V

VPROT

C35
100n 50V X7R

U3
TPS54302DDCR
VIN
EN
GND
BOOT
SW
FB

MID_BOOT
MID_SW
MID_FB

L1
6.8uH shielded CANDIDATE

MID

EF_ENABLED

R9
100k 0.1%

R10
12.9k 0.1%

C5
10u 25V X7R

C7
100n 50V X7R

C6
22u 25V X7R

C37
22u 25V X7R

GND

U5: P5 / 5.011 V

C36
100n 50V X7R

U5
TPS54302DDCR
VIN
EN
GND
BOOT
SW
FB

FIVE_BOOT
FIVE_SW
FIVE_FB

L4
6.8uH shielded CANDIDATE

P5

R17
100k 1%

R18
13.5k 0.1%

C18
10u 25V X7R

C20
100n 50V X7R

C19
22u 25V X7R

C38
22u 25V X7R

GND

BOOT caps are switch-node referenced, not grounded.
Kelvin FB ground to IC GND; keep high dI/dt loops off analog return.
6.8uH magnetics here are explicit candidates, not qualified choices.

INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE

Big Modular

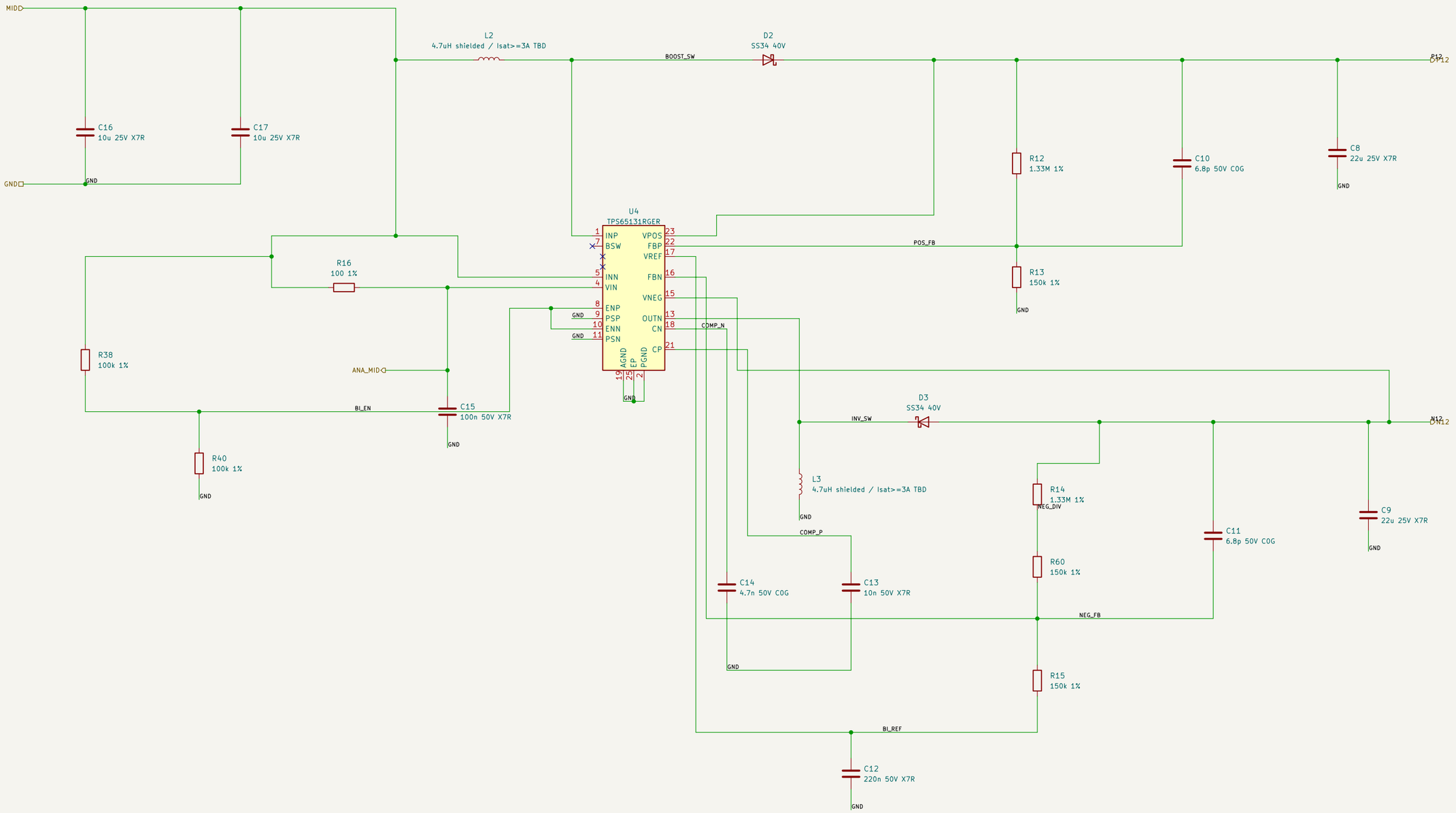
Sheet: /Independent buck converters/
File: 03_bucks.kicad_sch

Title: Big Modular — Independent buck converters

Size: User	Date:	**Rev: Wired review**
KiCad E.D.A. 10.0.6		Id: 4/8

04 / DIRECT BIPOLAR RAILS

Nominal +/−11.968 V. FBN returns to VREF, NOT GND; PSP/PSN grounded.
Local output feedback: no remote sense or arbitrary LC added. No rail LDOs.
-800 mA positive is chart interpolation at 25 C, NOT guaranteed continuous or simultaneous +/−800 mA.



Separate INP/INN local ceramic loops; shielded L2/L3.
No arbitrary output LC. Compensation and Cff require loop validation.

INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE

Big Modular

Sheet: /Direct bipolar rails/
File: 04_bipolar.kicad_sch

Title: Big Modular — Direct bipolar rails

Size: User	Date:	**Rev: Wired review**
KiCad E.D.A. 10.0.6		Id: 5/8

05 / RAIL DEBUG AND SINGLE POWER LED

J8 KEYED 2x10 DEBUG ONLY — NOT EURORACK POWER. No onboard MCU / no blink. D10 is ordinary +5V power indication.
3.3 V external controller: inputs HI-Z, outputs OPEN-DRAIN ONLY; no external pullups/power into AUX.
Normal operation is hardware PG/window only; qualification JP1 stays OPEN. See docs/debug-interface.md for all pins and sequencing.

ONE RED D10, HAND SOLDERED / ONE PANEL HOLE
Final +5V indication only; no blink or all-rail health claim.

J8: OPTIONAL 20-PIN SENSING / DEBUG — NOT POWER
Pin1 = GND / red stripe;17,20 explicitly NC.
AUX_MON through10k is monitor ONLY, not controller power.

R32
1k 1%
P5
LED_P_A
D10
RED 3mm / +5V POWER / MANUAL
GND

R61
10k 1%
AUX3V3

GND
J8
KEYED 2x10 DEBUG — NOT POWER
1 2
3 4
5 6
7 8
9 10
11 12
13 14
15 16
17 18
19 20
DBG_AUX_MON
ADC_P12
GND
ADC_N12
ADC_P5
ADC_MID
ADC_VBUS
ADC_VPROT
GND
GND

R62
10k 1%
PD_PG_ND
DBG_PG_N

DBG_FLT_N
R63
10k 1%
EF_FLT_N

AUX3V3
R70
10k 1%
R64
1k 1%
PD_SDA
DBG_SDA

AUX3V3
R71
10k 1%
R65
1k 1%
DBG_SCL
PD_SCL

R66
10k 1%
BUTTON_N
DBG_BUTTON_N

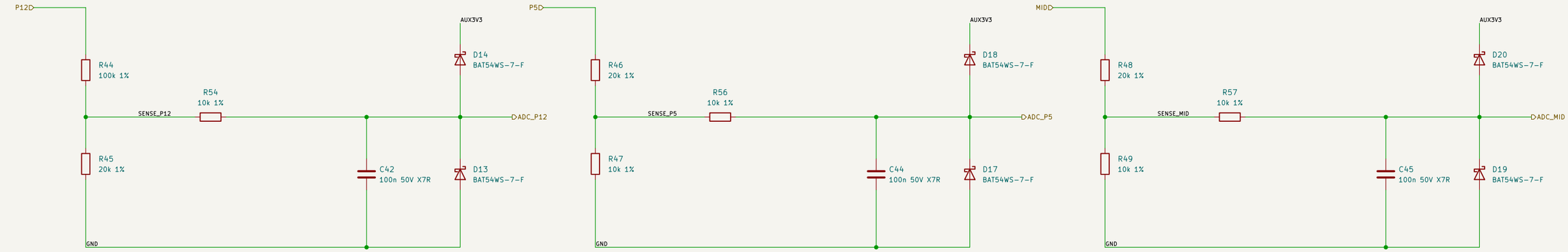
AUX3V3
R41
10k 1%
SW1
TL3340AF160QG
1 COMMON
3 COMMON
2 NO
CASE MP
C40
100n 50V X7R
GND

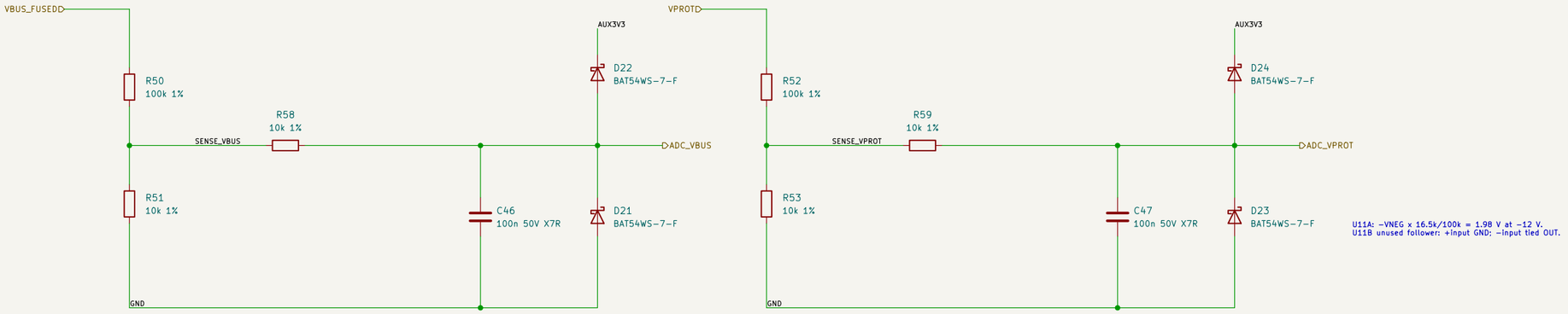
Local10k pullups only. External I2C must be OPEN DRAIN.
No external supply or pullups; no external enable dependency.
SW1 is a debug input, NOT a power bypass. JP1 stays OPEN for qualification.

INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE
Big Modular
Sheet: /Rail debug and single power LED/
File: 05_diagnostics.kicad_sch
Title: Big Modular — Rail debug and single power LED
Size: User | Date: | **Rev: Wired review**
KiCad E.D.A. 10.0.6 | Id: 6/8

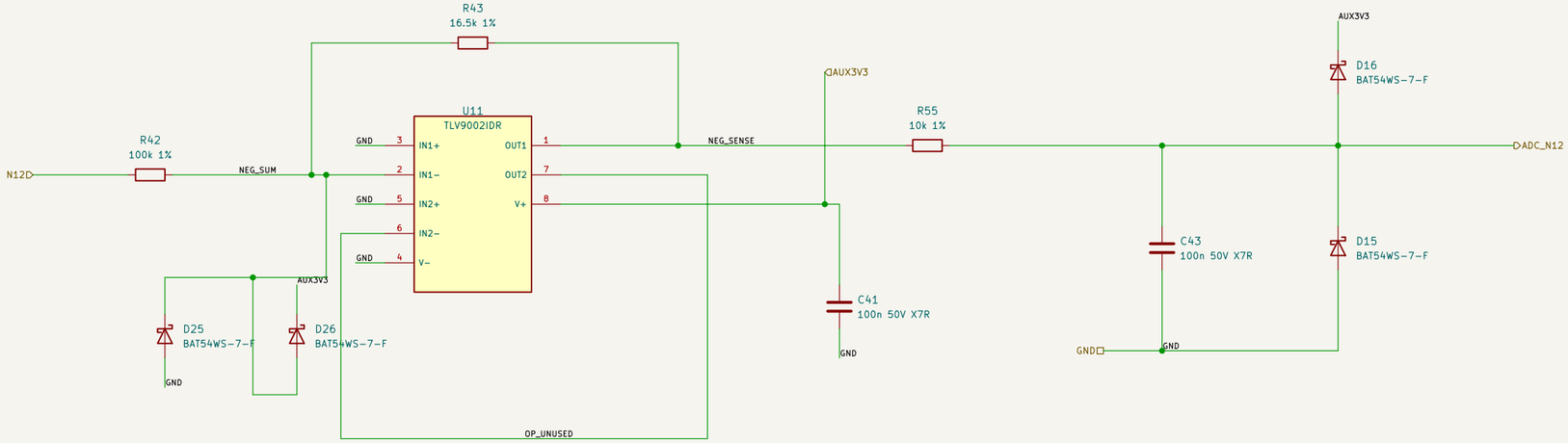
06 / RAIL SENSING AND INPUT PROTECTION

Negative rail uses ground-referenced inverter, NOT AUX-biased divider.
External Schottky clamps + series resistors are provisions, NOT proven powered-off isolation.
OFF-state injection / clamp Vf / residual rail bleed / reference calibration are POWER-UP BLOCKERS.





U11A: −VNEG x 16.5k/100k = 1.98 V at −12 V.
U11B unused follower: +Input GND; −Input tied OUT.



INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE

Big Modular

Sheet: /Rail sensing and input protection/
File: 06_sensing.kicad_sch

Title: Big Modular — Rail sensing and input protection

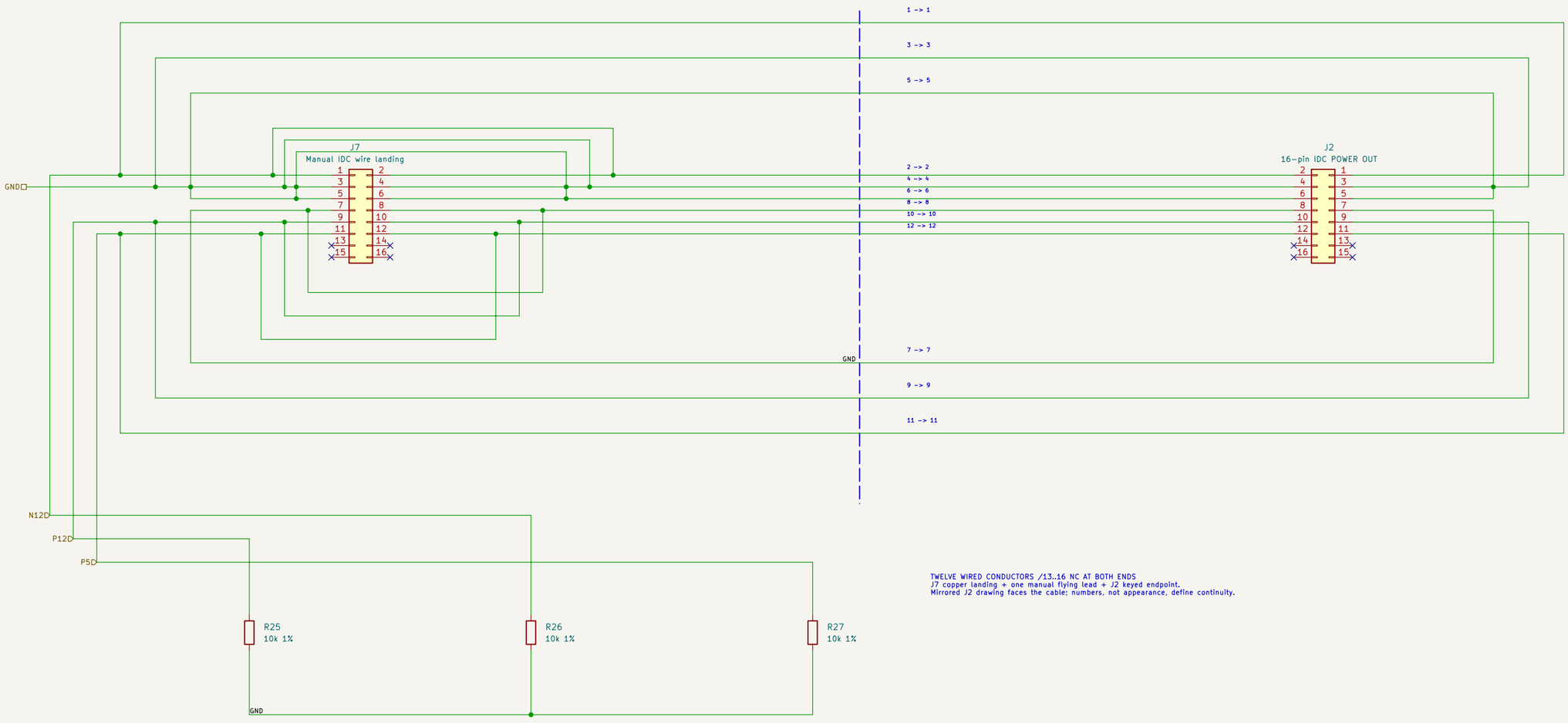
Size: User	Date:	**Rev: Wired review**
KiCad E.D.A. 10.0.6		Id: 7/8

07 / MANUAL OUTPUT HARNESS

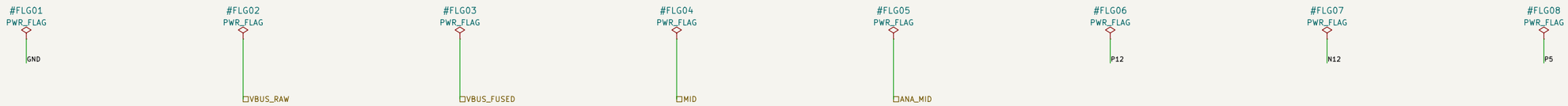
J7 board wire landing -> J2 off-board IDC cable, pin-for-pin.
1/2=-12, 3..8=GND, 9/10=+12, 11/12=+5; 13..16 NC.
Never connect another active PSU. Continuity-test red stripe / orientation before use.

J7: PCB WIRE LANDING

J2: OFF-BOARD KEYED ENDPOINT



TWELVE WIRED CONDUCTORS /13..16 NC AT BOTH ENDS
J7 copper landing + one manual flying lead + J2 keyed endpoint.
Mirrored J2 drawing faces the cable; numbers, not appearance, define continuity.



Independent ERC power declarations only — NOT additional physical supplies.
Passive flags retain the existing circuit; never parallel another active rack PSU.

INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE

Big Modular

Sheet: /Manual output harness/
File: 07_output_rails.kicad_sch

Title: Big Modular — Manual output harness

Size: User	Date:	**Rev: Wired review**
KiCad E.D.A. 10.0.6		Id: 8/8